

RICHARD GAO

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EDUCATION

Brown University

Sc.M. in Computer Science

Providence, RI

Aug. 2025 – May 2027

- Incoming student for Fall 2025 pursuing the Artificial Intelligence/Machine Learning thesis track.

UC SANTA BARBARA

B.S. in Computer Science • GPA: 3.95/4.0

Santa Barbara, CA

Sept. 2021 – Dec. 2024

- **Relevant Coursework (* = graduate-level):** Network Modeling and Graph Machine Learning*, Uncertainty Quantification and Scientific Machine Learning*, Deep Learning, Natural Language Processing, Machine Learning, Artificial Intelligence, Vector Calculus, Linear Algebra, Probability and Statistics, Discrete Math
- **Activities & Honors:** Dean's Honors (all quarters), Graduated with High Honors (top 8.5%), SB Hacks, Coders SB

EXPERIENCE

UCSB Computer Science

Researcher

Santa Barbara, CA

Feb 2025 – present

- Scaling Kolmogorov-Arnold physics-informed neural networks with Prof. Sanjukta Krishnagopal.
- Researching dynamic low-rank training through singular value decomposition of basis functions for solving partial differential equations.

UCSB Electrical & Computer Engineering

Researcher

Santa Barbara, CA

May 2024 – present

- Conducting research in efficient machine learning advised by Prof. Zheng Zhang.
- Researched meta-learning approaches to zeroth-order backpropagation-free training for adversarial attacks.
- Independently investigating task-agnostic knowledge distillation-based compression on LLMs.
- Independently developing a dynamic low-rank training algorithm for tensor-train decomposed linear layers.
- Collaborating with lab members to benchmark distillation methods to recover pre-training performance of low-rank adapted LLMs and Vision Transformers.
- Attend weekly seminars and lab meetings to present findings and discuss novel, efficient ML papers.

PROJECTS

KAN We Tensorize GraphSAGE

November 2024

- Explored Kolmogorov-Arnold Networks (KANs) as an alternative to multilayer perceptrons in GraphSAGE.
- Investigated compression using tensor-train decomposition on learnable weights of GraphSAGE.
- Implemented modified aggregators in PyTorch and performed experiments on protein-protein interactions, scientific paper network citations, and a synthetic dataset.
- Demonstrated the effectiveness of KAN layers and the low tensor rank of GraphSAGE weights.
- Completed a 10-page report detailing background, motivation, methods, and results.

ZERO-ORDER SPIKING NEURAL NETWORK TRAINING

May 2024

- Developed an end-to-end training framework for non-differentiable spiking neural networks via meta-learning for neuromorphic computing.
- Used local zeroth-order gradient with an LSTM meta-optimizer for variance reduction from conference papers.
- Implemented training algorithm in PyTorch and performed experiments on the MNIST dataset.
- Demonstrated transferability of the learned optimizer to different spiking neural network architectures.
- Achieved fast convergence and high accuracy against SoTA optimizers on a network of ~300k parameters.

MICROCYLINDER CLOTH RENDERER

March 2024

- C++ implementation of "A Practical Microcylinder Appearance Model for Cloth Rendering."
- Coded path-tracing with Russian roulette, bidirectional scattering distribution function, importance sampling, masking, and parallel processing.
- Demonstrated implementation by rendering different sets of parameters on a custom scene

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KNOWLEDGE NAVIGATOR

January 2024

- App for semantic document search with natural language queries on user-uploaded PDF files
- Used Google Cloud DocumentAI Optical Character Recognition API to identify text on uploaded documents
- Implemented a pre-trained transformer encoder-only network to obtain dense text and query embedding vectors
- Designed a reranking and cosine similarity-based comparison to display relevant text to the user

3D ENVIRONMENT GENERATOR

March 2023

- Web app to procedurally generate custom 3D environments through a graphical interface by uploading custom 3D building blocks; used React.js, Firebase, Flask, AWS EC2, MongoDB.
- Helped lead and facilitate sprint meetings and plan deliverables.
- Generates up to 20x20x20 block environments using the Wavefunction Collapse algorithm.
- Designed the block-adjacency labeling algorithm for mesh matching.
- Wrote Python implementation for automatic 3D block labeling and adjacency rule generation for the wavefunction collapse algorithm, enabling automatic constraints for user-uploaded meshes.

SKILLS & INTERESTS

Programming Languages: C++, C, Python, Java, JavaScript, SQL

Software: PyTorch, Jax, Numpy, MATLAB

Languages: English (fluent), Mandarin (fluent), Spanish (conversational)

Interests: Piano, botany, basketball, running